

Product Scope

Revenue Management Suite for Small Hotels in Indonesia

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Status: Planning — pre-development

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Target market: Hotels < 50 rooms · Indonesia

Stack: SvelteKit + Tailwind v4 · Go · PostgreSQL

Business model: SaaS monthly subscription

Primary users: Owner + 1-2 receptionists (roles: owner / receptionist)

This document defines the complete product scope, feature roadmap, data model, and technical architecture for TEREN Hotels — a focused revenue management tool designed to help small Indonesian hotels transition from paper-based operations to a clean, fast, and intentional digital workflow.

1. Product Vision

TEREN Hotels is built for the majority of Indonesian hospitality — guesthouses, losmen, family villas, and boutique properties in Bali, Lombok, Yogyakarta, and beyond. Most of these operate on paper, WhatsApp, or basic spreadsheets with no visibility into revenue performance, no pricing strategy, and no single source of truth for room availability.

The product does not compete with enterprise PMS solutions. It is the tool that comes before: the first digital step that feels as natural as the notebook it replaces.

Mission	Eliminate friction between a hotel's daily operations and its revenue intelligence.
Core promise	Fast to learn. Fast to use. No registration walls. No features without purpose.
Out of scope	OTA channel sync, external booking widget, consumables/minibar inventory, mobile app (Phase 1).
Pricing model	SaaS monthly subscription. No setup fees. No per-booking commissions.

2. Feature Roadmap

PHASE 1 — MVP CORE · Months 0–4

Module	Features
Property setup	Hotel profile (name, location, currency, timezone) · Room type configuration (name, max occupancy) · Floor builder with sort order · Room creation and numbering
Floor map & rooms	Visual floor map with drag-and-drop room positioning (pos_x / pos_y) · Room status at a glance (available, occupied, blocked, pending) · Room assignment from floor map when creating a booking
Inventory	Availability calculated in real time from booking ranges (check_in / check_out) · Manual room blocks: maintenance, owner use, out of service · Availability query per date range with conflict detection · Overbooking prevention at booking creation
Pricing engine	Base rate per room type · Weekend rate override · High/low season rates with date ranges · Minimum stay rules · Promotional discount codes
Reservation management	Create / edit / cancel booking · Guest profile (name, ID/passport, phone, nationality) · Check-in and check-out flow · Room assignment from floor map · Booking source tag (walk-in, WhatsApp, phone, Booking.com, Airbnb, Agoda, Traveloka, other)
Invoicing & payments	Invoice generated per booking · PDF export · Payment methods: cash, bank transfer, QRIS, card · Payment status: pending / partial / paid · Daily cash summary report

PHASE 2 — REVENUE INTELLIGENCE · Months 4–8

Module	Features
Revenue dashboard	Daily / weekly / monthly RevPAR · Occupancy rate chart · ADR (average daily rate) tracker · Revenue by room type · Income vs target goal widget
Guest communication	WhatsApp booking confirmation template · Pre-arrival message · Post-stay review request · Guest history and notes
Reporting	Tax report (PPN, Indonesian compliance) · Monthly P&L; export (CSV / PDF) · Booking source attribution · Forecast: next 30 / 60 / 90 days

PHASE 3 — GROWTH & SCALE · Months 8–14

Module	Features
Dynamic pricing engine	Demand-based auto price suggestions · Local holiday detection (Indonesia) · Competitor rate benchmarking · Accept / reject pricing nudges
Multi-property	Manage 2–5 properties under one account · Portfolio revenue overview · Staff roles per property
Direct booking page	Embeddable booking form · Custom booking page (teren.id/hotel-slug) · Real-time availability display · Online payment

3. Key Revenue Metrics (tracked from day 1)

Code	Metric	Formula
RevPAR	Revenue Per Available Room	Total revenue ÷ total available rooms
ADR	Average Daily Rate	Total room revenue ÷ rooms sold
Occupancy %	Occupancy Rate	Rooms sold ÷ total rooms × 100
Source mix	Booking Channel Split	Direct % vs OTA % vs walk-in %

4. Data Model

The schema follows Clean Architecture principles: no business logic in the database, row-level security per property from day one (multi-property ready), and no redundant state — inventory availability is always computed from bookings and blocks, never stored.

Entity	Key fields
PROPERTY	id · name · slug · currency · timezone · settings (jsonb)
USER	id · property_id · name · email · password_hash · role (enum) · is_active
FLOOR	id · property_id · floor_number · label · sort_order
ROOM_TYPE	id · property_id · name · max_occupancy · description
ROOM	id · floor_id · room_type_id · number · status (enum) · pos_x · pos_y
RATE_RULE	id · room_type_id · base_rate · weekend_rate · high_season_rate · season_start · season_end · min_stay_nights
ROOM_BLOCK	id · room_id · created_by · start_date · end_date · reason (enum) · notes
GUEST	id · property_id · full_name · id_number · phone · email · nationality · notes
BOOKING	id · property_id · room_id · guest_id · created_by · check_in · check_out · adults · children · total_amount · source (enum) · status (enum) · notes
INVOICE	id · booking_id · invoice_number · subtotal · tax_amount · total · payment_method (enum) · payment_status (enum) · issued_at · paid_at

4.1 Inventory logic

Inventory is not a table — it is a derived state. A room is considered available for a date range if no active booking overlaps that range AND no room block overlaps that range. The availability query uses a NOT IN / NOT EXISTS pattern against BOOKING (status != cancelled) and ROOM_BLOCK, joined on date overlap: check_in < end_date AND check_out > start_date.

4.2 Enumerations

Enum	Values
USER.role	owner · receptionist
ROOM.status	active · inactive
ROOM_BLOCK.reason	maintenance · owner_use · out_of_service
BOOKING.status	confirmed · checked_in · checked_out · cancelled · no_show
BOOKING.source	walk_in · whatsapp · phone · booking_com · airbnb · agoda · traveloka · other
INVOICE.payment_method	cash · bank_transfer · qris · card
INVOICE.payment_status	pending · partial · paid

5. Technical Architecture

Frontend	SvelteKit + Tailwind v4. Web-first. No mobile app in Phase 1. Offline-first mindset — critical data cached locally.
Backend	Go (Golang) — Clean Architecture Light. Handlers → Services → Repositories. No ORM — raw SQL via pgx driver.
Database	PostgreSQL. Migrations via golang-migrate. Row-level security per property_id from day one.
Auth	JWT tokens. Two roles: owner (full access) and receptionist (bookings, check-in/out, invoices — no pricing config, no financial dashboard).
PDF generation	Server-side PDF generation for invoices (Go PDF library). Stored in S3-compatible storage (Cloudflare R2 — zero egress cost).
Infra / deploy	Docker Compose locally. Fly.io or Railway for production — Singapore region for low latency to Indonesia. Cloudflare CDN for static assets.
Flutter / mobile	Backlog only. No mobile app until explicitly scoped. Web app is fully responsive and works on tablet at reception desk.

5.1 API structure (Go)

Service	Responsibility
Auth handler	Login, JWT issue/refresh, role enforcement
Property service	Property CRUD, floors, room types, rooms, pos_x/pos_y
Inventory service	Availability query, room block CRUD, conflict detection
Booking service	Booking CRUD, check-in/out flow, rate calculation
Pricing service	Rate rule CRUD, rate resolution for a given date range
Invoice service	Invoice generation, PDF export, payment status updates
Report service	RevPAR, ADR, occupancy queries, source attribution

6. Design Principles (TEREN brand applied)

Outdoor-first	High contrast palette. Warm Stone background (#F5F4F1). Deep Stone text (#1C1917). Usable in direct sunlight on a tablet at a Bali guesthouse reception.
Inline editing	No modal forms for common actions. Prices, room names, and statuses are edited inline — one tap, change, done.
Guest-first flow	Staff can create a booking in under 30 seconds. No required fields beyond room, guest name, and dates.
Never block the user	Overbooking warnings are warnings, not hard blocks (owner can override). Minimum stay violations surface as alerts, not errors.
Fluid micro-animations	Room status transitions, availability calendar updates, and KPI counters animate smoothly (tabular-nums, tweening).
Whitespace & hierarchy	Cards nest on a warm stone canvas. Lighter card surface (#FCFBFA) over the page background (#F5F4F1). Rounded corners 8–16px throughout.

Flow systems. Soulful experiences.

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